

Introduction to Modern AI, Transformers and Agentic solutions



יוסי זגורי

Yossi@integrative.net

052-4668866

Topics:

AI

DL

TRANSFORMERS

LLM

OLLAMA

RAG



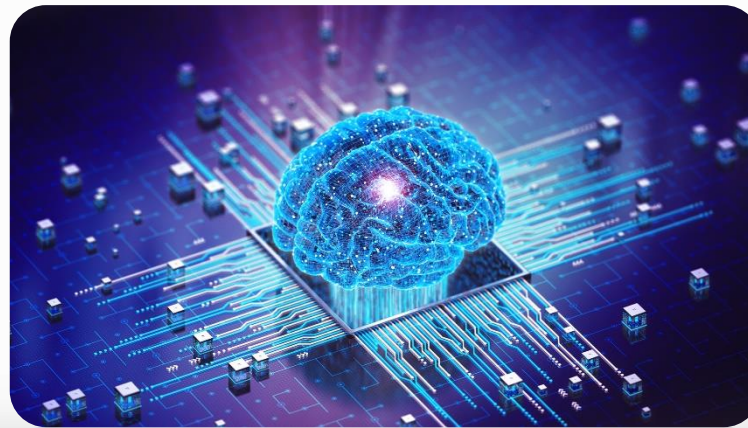
AI Background

Cognition, Intelligence & AI

Cognition refers to the human mental processes involved in acquiring knowledge and understanding through thought, experience, and senses, using perception, memory, attention, reasoning, problem-solving, and decision-making

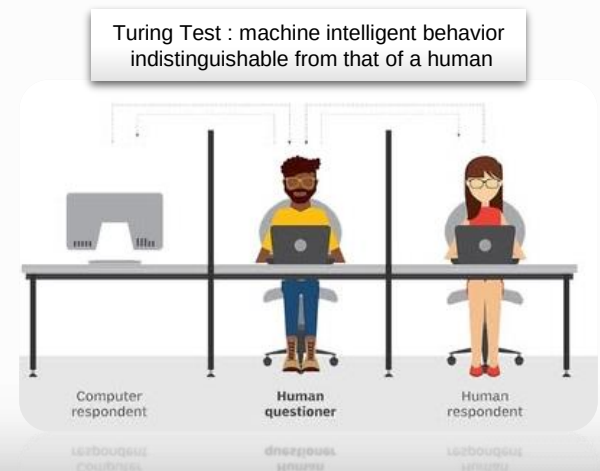
Intelligence is often considered the ability to use these cognitive processes effectively, i.e. applying abstraction, logic, understanding, self-awareness, learning, emotional knowledge, reasoning, planning, creativity, critical thinking, and problem-solving resulting adaptation to new situations, learning from experience, and solving problems.

Artificial intelligence (AI) is a set of technologies enabling computers and machines to learn from the environment, comprehend, perform problem solving, decision making, display creativity and autonomy, seemingly behaving/solving complex tasks as humans do.



Milestones in AI progress

- **Early Foundations (1950s):**
 - Alan Turing proposed the concept of machine intelligence and introduced the Turing Test (1950).
 - Early AI programs like the "Logic Theorist" focused on problem-solving and symbolic reasoning (FOL).
- **Development of Expert Systems (1970s-1980s):**
 - Expert systems like DENDRAL, INTERNIST-1 and MYCIN simulated human expertise in specific domains.
 - Inference Engines, Rule-Based Systems, Human-Machine Q&A
 - These systems were used for tasks such as medical diagnosis and chemical analysis.
- **Rise of Connectionism (Late 20th Century):**
 - Emphasis on artificial neural networks (ANN) to mimic human brain functionality.
 - Advancements in machine learning and pattern recognition.
- **Modern AI Advancements (21st Century):**
 - Applications include natural language processing, robotics, image recognition, and autonomous systems.
 - Breakthroughs in deep learning and data-driven AI models.

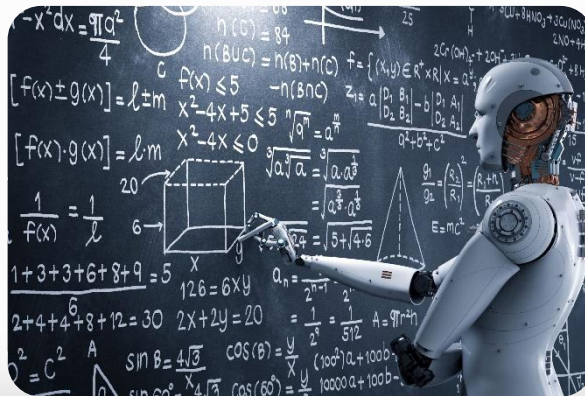


AI Types

Narrow AI (Weak AI) : Type of AI designed and trained to perform a specific task or a narrow range of tasks. It doesn't possess general intelligence and operates within predefined parameters. Examples: Virtual assistants (like Siri or Alexa), recommendation systems, facial recognition software, and self-driving car systems.

General AI (Strong AI): Also referred as AGI, Hypothetical at this point, representing a machine that has the ability to understand, learn, and apply intelligence across a wide range of tasks, similar to human cognitive abilities. It can adapt and transfer knowledge across domains. Currently, there are no true examples of General AI, though advancements in AI research continue to move in this direction (Open AI O3)

Superintelligent AI: A level of AI that surpasses human intelligence in all areas, including creativity, problem-solving, and decision-making. It could outperform the best human minds in every field.



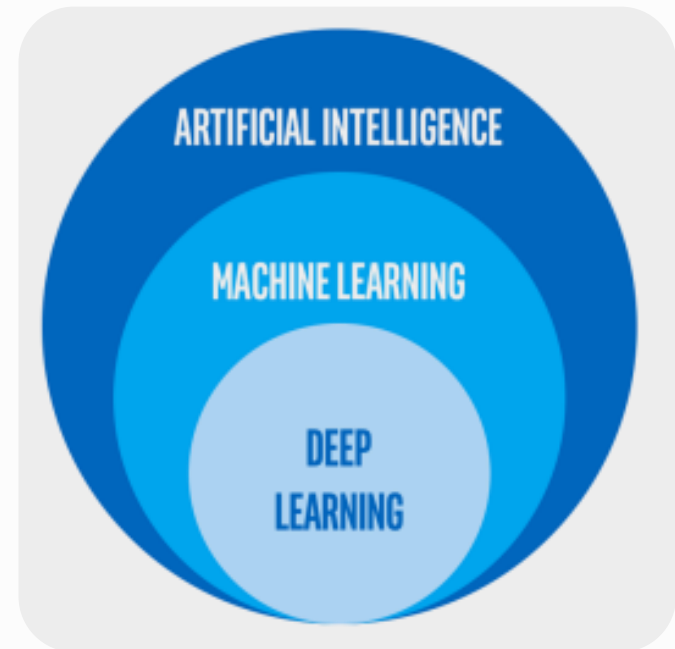
- CHAIN OF THOUGHT - Prompt engineering – Wikipedia הרעיון של

Learning in ML

Learning is **Improving** with **Experience** at some **Task**
With respect to some performance Measure

Machine learning is a subset of AI which is incorporated into Devices or software applications - enabling them to be more precise & accurate at finding or predicting outcomes

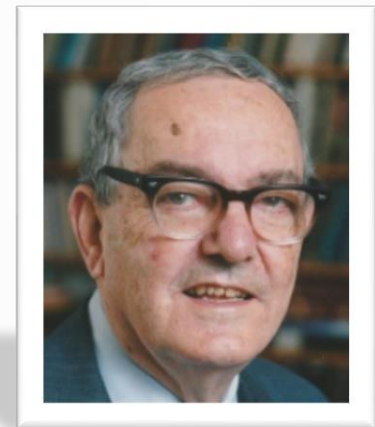
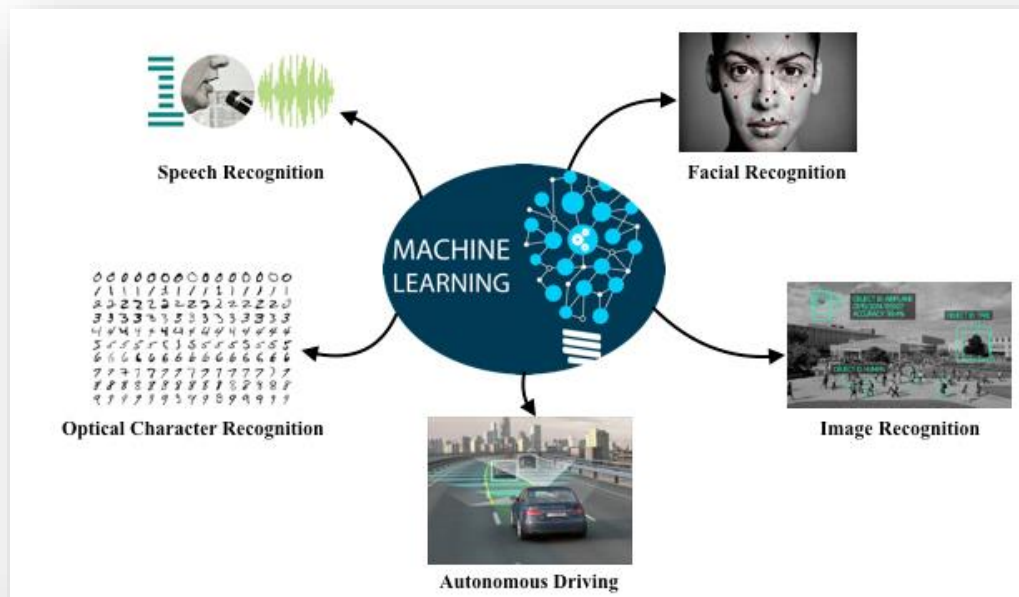
ML used to analyze data and discover patterns, correlations or rules hiding within the data



Machine Learning

“Machine Learning is concerned with computer programs that automatically improve their performance through experience”

“Learning is any process by which a system improves performance from experience”



Herbert Alexander Simon
(Nobel Prize Economics 1978)

Machine Learning

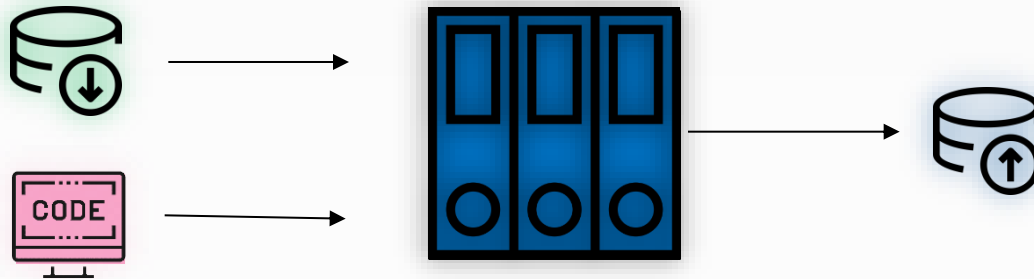
“Machine learning is a method of data analysis that automates model building. Using algorithms that iteratively learn from data.

Machine learning allows computers to find hidden insights without being explicitly programmed where to look.”

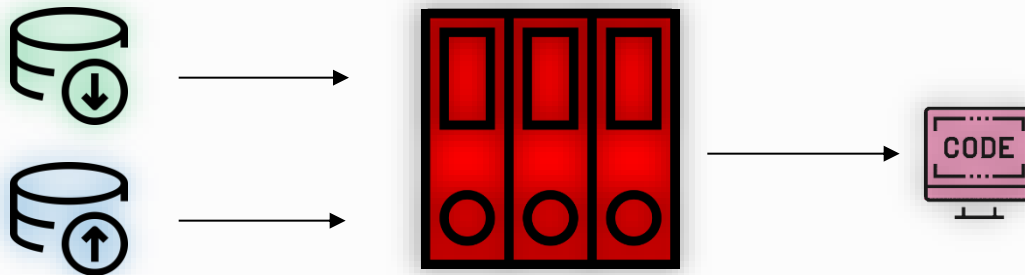
Source: <http://www.sas.com/>

Machine Learn in a nutshell

20' Centaury (programming)



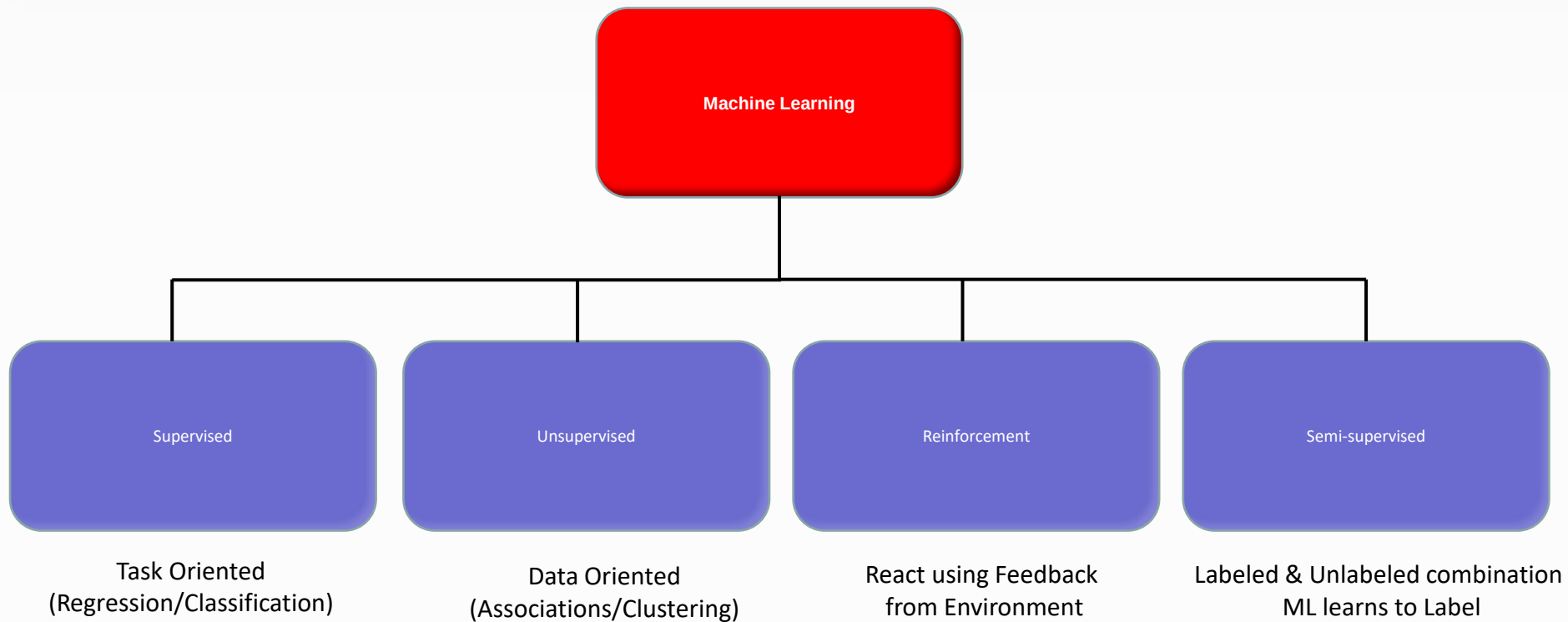
21' Centaury (machine learning)



“For 2019, ML and AI will finally find their way out of the lab and into existing applications one way or another. For the most part, people won’t even know it’s there because it will be embedded in a seamless fashion.” [_Scott Parker, Director, Sinequa.](#)

<https://dzone.com/articles/aiml-2018-surprises-and-2019-predictions>

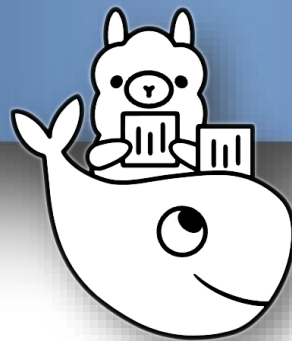
Types Of Machine Learning



Demo

[What is Deep Learning](#)

[Transformers in Machine Learning - GeeksforGeeks](#)

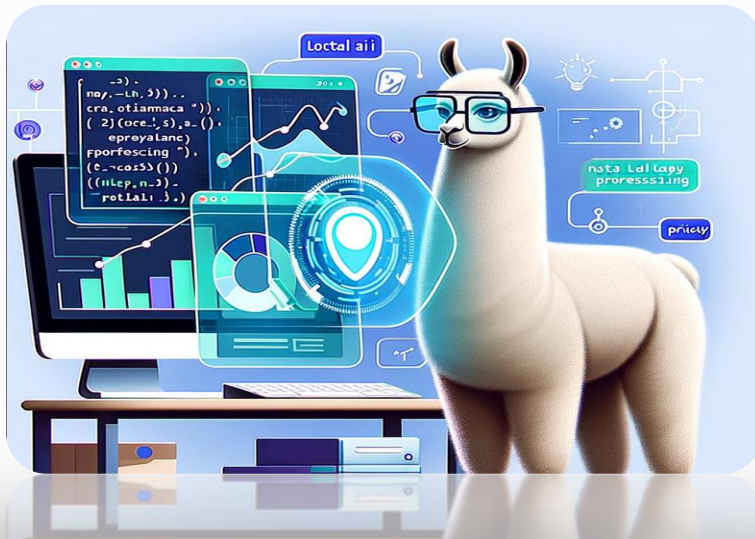


Ollama



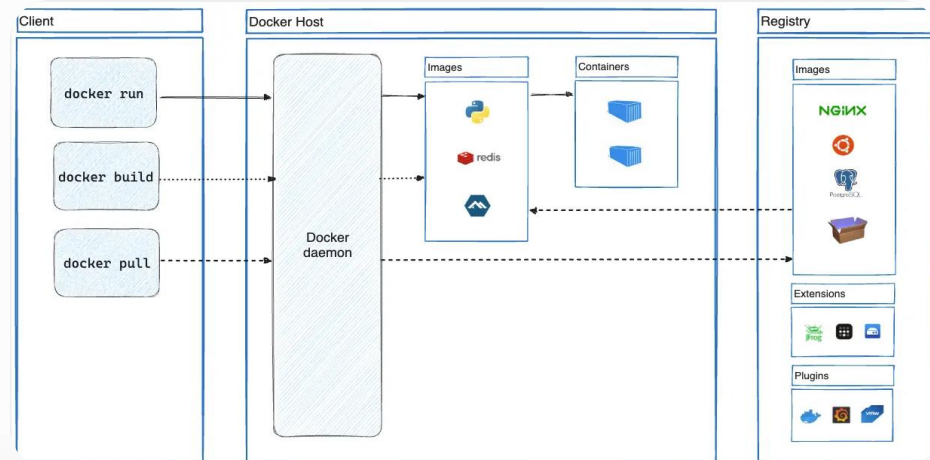
What is Ollama and Docker

- 🌀 Ollama enables running **large language models (LLM)** locally
- 🌀 Ollama can act as your *personal* LLM Advisor
- 🌀 Local Installation: [Ollama Tutorial: Running LLMs Locally Made Super Simple - KDnuggets](#)
- 🌀 Select LLM Model : [library](#)



What is Ollama and Docker

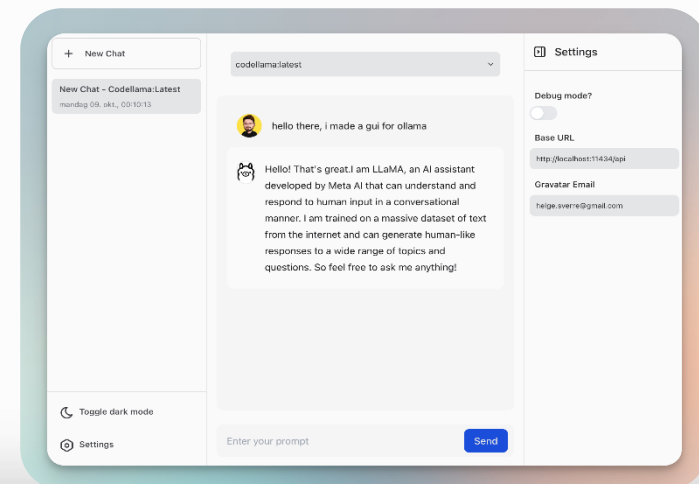
- 🌀 Ollama enables running large language models locally
- 🌀 Ollama can act as your *personal* LLM Advisor
- 🌀 Docker is a *container*, lightweight virtual machine, open platform for developing, shipping, and running applications, enabling separation & isolation of applications environment from the infrastructure.
- 🌀 Together, they create good setup for hassle-free LLM exploration.



[What is Docker? | Docker Docs](#)

Setup Stages

- Download and install *Docker Desktop* (Windows and macOS) or Docker Engine (Linux).
- Choose model from the [Ollama library](#) (tinyllama, mistral, llama3).
- Download the Ollama Docker image: One simple command (docker pull ollama/ollama) gives you access to the magic.
- Run the Ollama container: Customize for CPU/GPU
- Navigate to <http://localhost:11434> for Ollama's Web GUI
- What can be done ?
 - Send prompts & receive creative responses.
 - Fine-tune your model on your own data.
 - Explore a variety LLM tasks: Translation, question-answering, code completion etc.



Ollama Docker Image

- Download and install : [Windows | Docker Docs](#)
- Install and run Ollama Image (cpu version):

```
docker run -d -v /d/ollamadata:/root/.ollama -p 11434:11434 --name ollama ollama/ollama
```

- docker run:** Basic command used to start a new container from a Docker image. If the image is not already downloaded, it will be pulled from Docker Hub.
- d:** "detached mode". runs the container in the background, allowing you to continue using your terminal without being attached to the container's process.
- v :** mounts D:\ollamadata to /root/.ollama inside the container, ensuring any data in that folder on your Windows machine will persist and be accessible by the container.
- p 11434:11434:** maps container's internal port 11434 to port 11434 on host machine.
- name ollama ollama:** assigns the name ollama to the running container.
- ollama/:** The name of the Docker image being used to create the container. In this case ollama/ollama image, a pre-built image stored in a Docker registry.

<http://localhost:11434/>

For Mac or GPU read: [How to Install and Run Ollama with Docker: A Beginner's Guide – Collabnix](#)

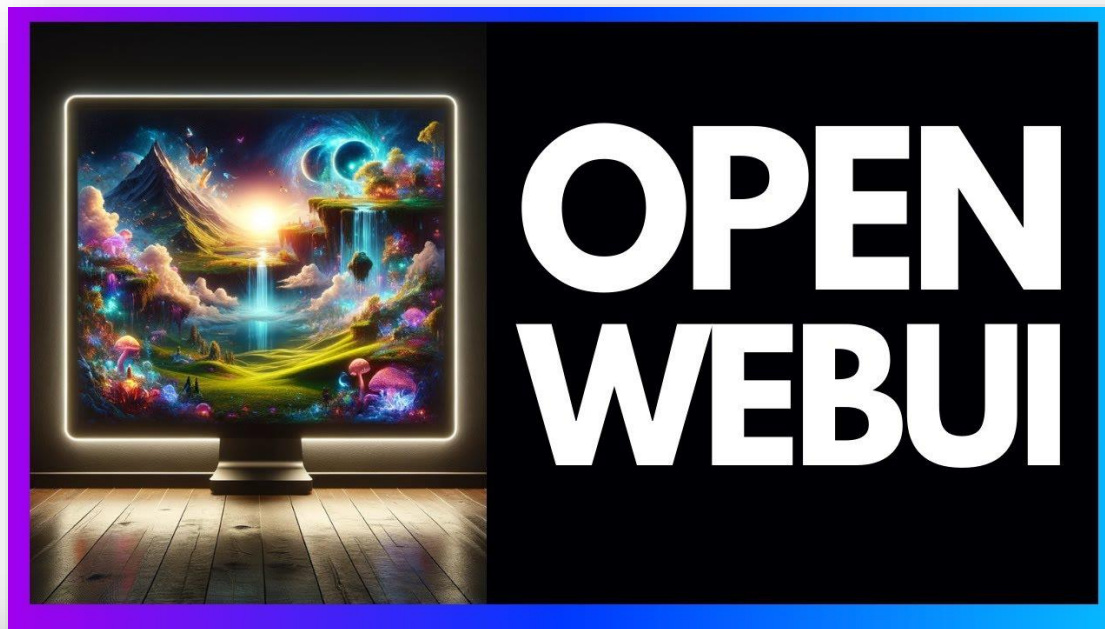
Ollama Docker Image

- Download and Run model (tinyllama or mistral) in CLI

```
docker exec -it ollama ollama run tinyllama
```

- Or install open-webui for GUI:

[open-webui/open-webui: User-friendly AI Interface \(Supports Ollama, OpenAI API, ...\)](#)



Docker CLI



CLI Cheat Sheet



Docker provides the ability to package and run an application in a loosely isolated environment called a container. The isolation and security allows you to run many containers simultaneously on a given host. Containers are lightweight and contain everything needed to run the application, so you do not need to rely on what is currently installed on the host. You can easily share containers while you work, and be sure that everyone you share with gets the same container that works in the same way.

INSTALLATION

Docker Desktop is available for Mac, Linux and Windows
<https://docs.docker.com/desktop>

View example projects that use Docker
<https://github.com/docker/awesome-compose>

Check out our docs for information on using Docker
<https://docs.docker.com>

IMAGES

Docker images are a lightweight, standalone, executable package of software that includes everything needed to run an application: code, runtime, system tools, system libraries and settings.

```
Build an Image from a Dockerfile
docker build -t <image_name> .
```

```
Build an Image from a Dockerfile without the cache
docker build -t <image_name> . --no-cache
```

```
List local images
```

GENERAL COMMANDS

```
Start the docker daemon
docker -d
```

```
Get help with Docker. Can also use --help on all subcommands
docker --help
```

```
Display system-wide information
docker info
```

CONTAINERS

A container is a runtime instance of a docker image. A container will always run the same, regardless of the infrastructure. Containers isolate software from its environment and ensure that it works uniformly despite differences for instance between development and staging.

```
Create and run a container from an image, with a custom name:
docker run --name <container_name> <image_name>
```

```
Run a container with and publish a container's port(s) to the host.
```

Connect Ollama using Python

- Create project (folder, virtual environment, activate it)
- pip install ollama

```
import ollama

response = ollama.generate(model='tinyllama',
prompt='what is a model?')

print(response['response'])
```

Connect Ollama using Python

```
docker exec -it ollama ollama run llama2
```

🌐 Using Langchain

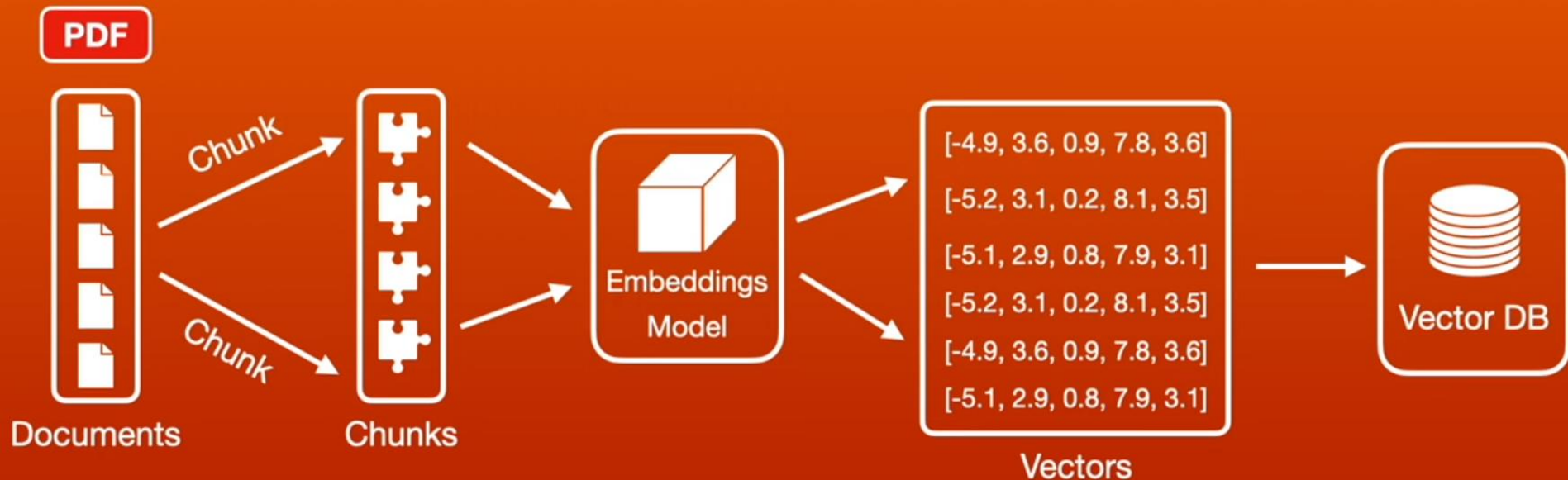
```
from langchain_community.llms import Ollama  
  
llm = Ollama(model="llama2")  
  
llm.invoke("tell me about partial functions in python")
```



RAG with Ollama

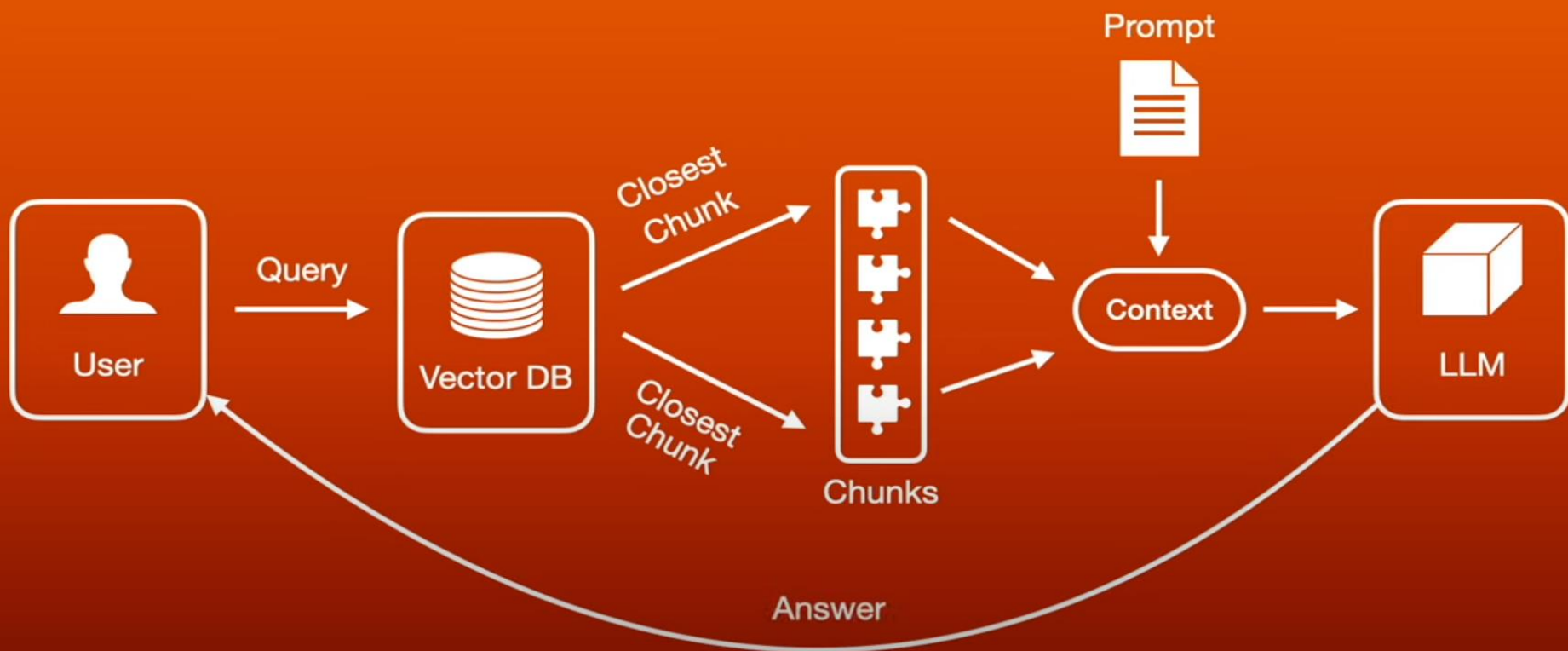
RAG Development

Phase One: Ingestion



RAG Development

Phase Two: Query



RAG Development

Tools



Python



LangChain



Ollama



Llama 3.2



Chroma



Langchain

What is Langchain

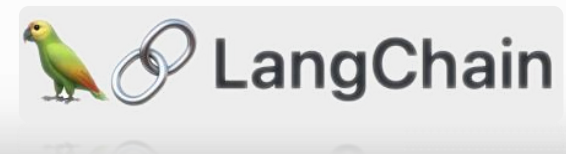
- Framework for building applications based on large language models (LLMs) that generate *new content* (answering questions, create image from text prompts etc.)
- Enables the use of pre-trained models, provides tools and abstractions to improve the customization, accuracy, and relevancy of the information the models generate.
- LangChain streamlines steps to develop data-responsive applications, diverse applications powered by language models more effortlessly, including chatbots, question-answering, content generation, summarizers, and more.
- LangChain enable repurposing LLMs for *domain-specific* applications without retraining or fine-tuning (which may be very expensive).
- LangChain facilitates developing **Retrieval Augmented Generation (RAG)** workflow that introduces new information to the language model during prompting.
- Implementing context-aware workflows like RAG reduces model hallucination and improves response accuracy.



What is Langchain

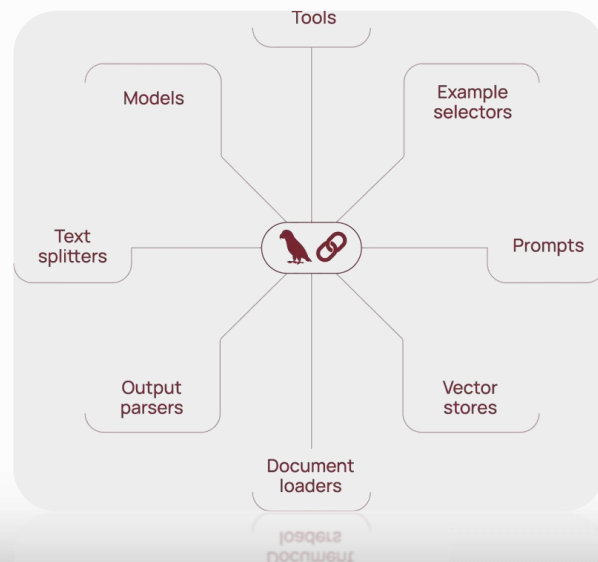
- LangChain connects LLMs company's private data and APIs to build context-aware, reasoning applications, like **RAG** or simple chains.
- LangChain implements a standard interface for LLM's and related technologies, such as embedding models and vector stores, and integrates with hundreds of providers.
- LangChain is a Python framework that helps building AI applications and simplify all the requirements without having to code all the little details
- Examples of LangChain applications:
 - RAG
 - ChatBot
 - Synthetic data generation
 - Interacting with a database using NLP

[What Is LangChain?. By David Dieruf | by DataStax | Medium](#)



LangChain Features and Libs

- **langchain-core:** Base abstractions and LangChain Expression Language (LCEL) for composing Chains.
- **langchain-community:** Third-party integrations and partner packages for extensibility such as langchain-openai , lanchain-anthropic , etc.
- **Langchain:** Chains, Agents, and retrieval strategies for cognitive architecture.
- **LangGraph:** multi-actor applications with LLMs by modeling steps as edges and nodes in a graph.
- **LangServe:** Deploy LangChain Chains as REST APIs.
- **LangSmith:** Platform to debug, test, evaluate, and monitor LLM applications.



Demo

https://dev.to/mohsin_rashid_13537f11a91/rag-with-ollama-1049

From **automating complex tasks** to **analyzing vast datasets** and **generating intelligent content**,
AI agents are leading a tech revolution- and now, you have the chance to be a part of it.